

Notes on The Student Seminar on Modular Forms and Elliptic Curves

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This will be the notes for the Seminar on Elliptic Curves and Modular Forms during the fall semester of 2026 at the University of Maryland. This seminar is an introductory level treatment on the topics of elliptic curves and modular forms, following Neal Koblitz's book [Kob93] *Introduction to Modular Forms and elliptic curves*. We will also provide several write-ups on the algebraic backgrounds mainly following [Vak25].

1 Chapter 1: From Congruent Number to Elliptic Curves

1.1 Congruent Numbers

1.2 A Certain Cubic Equation

1.3 Elliptic Curves

Below are the exercises of this section.

Proposition 1.1. Suppose that K is an infinite field, and $F[x, y, z] \in K[x, y, z]$ is a polynomial of total degree n . Suppose F satisfies $F(\lambda x, \lambda y, \lambda z) = \lambda^n F(x, y, z)$ for all $(x, y, z) \in K^*$, then F is homogeneous.

Proof. Since F has total degree n , let's try to write F as a sum of all its lower degree homogeneous part:

$$F(x, y, z) = \sum_{d=0}^n F_d(x, y, z)$$

where $F_d(x, y, z)$ is the d -th homogeneous part of F , which is the summation of all the degree d monomials in F . Then by our assumption, we know that:

$$F(\lambda x, \lambda y, \lambda z) = \lambda^n F(x, y, z) = \sum_{d=0}^n \lambda^d \lambda^d F_d(x, y, z)$$

Then if we fix a $(x, y, z) \in K^3$ and consider the equation with respect to λ , we get:

$$\sum_{d=0}^n \lambda^d \lambda^d F_d(x, y, z) - \lambda^n F(x, y, z) = \sum_{d=0}^{n-1} \lambda^d \lambda^d F_d(x, y, z) + \lambda^n (F_n(x, y, z) - F(x, y, z)) = 0$$

Notice that this equation holds for every $\lambda \in K$, then we know that since K is infinite, this means that this polynomial is the 0 polynomial no matter what (x, y, z) we choose. In particular, we know that:

$$F_d(x, y, z) = 0, \quad 0 \leq d \leq n-1, \quad \forall (x, y, z) \in K^3$$

Again, since K is infinite, this implies that $F_d(x, y, z)$ is the 0 polynomial, therefore F only has the degree n part. $\square$

Proposition 1.2. Consider the projective space $\mathbb{P}_k^2$, here by a **line** in the projective space, we either mean the projective completion of a line of the ordinary xy -plane or the line at infinity. Then the equation of a line in $\mathbb{P}_k^2$ is of the form $ax + by + cz = 0$ where $a, b, c \in k$ are not all 0. Moreover, two such equations determines the same line iff two triples (a, b, c) differ by a multiple.

Proof. Recall that a line in the xy -plane is given by $ax + by + c = 0$ where a, b, c are not all 0, then we know that the projective completion of this line is given by:

$$ax + by + cz = 0, \quad a, b, c \in k$$

Now let's consider the line at infinite, which should consist of all the equivalence classes of the form $[x : y : 0]$, and we know that the equation for this line at infinity is given by:

$$z = 0$$

Notice that this is actually a special case of the previous equation where $a, b = 0$ while $c = 1$.

Then suppose that we have two lines, $ax + by + cz = 0$ and $a'x + b'y + c'z = 0$ and they are the same. This means that $[x : y : z]$ is on the first iff it's on the second. First notice that if $a \neq 0$, then $a' \neq 0$. To see why, if $a \neq 0$ but $a' = 0$, then we know that $[1 : 0 : 0]$ is on the second line but it's not the one first line. Similarly, we can assume that if $b = 0$ iff $b' = 0$ and $c = 0$ iff $c' = 0$. Then let's just assume that a, b, c are all nonzero, and we can assume that $\frac{a'}{a} = \lambda_a$, $\frac{b'}{b} = \lambda_b$ and $\frac{c'}{c} = \lambda_c$, which means that the equation of the second line is:

$$\lambda_a ax + \lambda_b bx + \lambda_c cx = 0$$

We want to show that $\lambda_a = \lambda_b = \lambda_c$. Let's first set $z = 0$, then we know that, for each $y \in K$, there is a unique $x \in K$ such that:

$$ax + by = 0 \Leftrightarrow ax = -by$$

Plug this into the equation of the second line, we get:

$$by(\lambda_b - \lambda_a) = 0, \quad \forall y \in K$$

This implies that $\lambda_a = \lambda_b$. Similarly we can prove $\lambda_c = \lambda_b$, thus we are done. $\square$

For the projective plane $\mathbb{P}_k^2$, there is a very useful **dual construction**, which we will show in the following exercise:

Proposition 1.3. Let's consider two copies of $\mathbb{P}_k^2$, one with coordinates x, y, z and another one with coordinates a, b, c . There is a one-to-one correspondence between lines in a copy of $\mathbb{P}_k^2$ with coordinates x, y, z and points in another copy with coordinates a, b, c . There is also a one-to-one correspondence between lines in the abc projective plane and points in the xyz projective plane. Under this correspondence, suppose that $p_1, \dots, p_n$ is a set of points in xyz plane, then they line in the same line iff the lines corresponding to them in the abc plane meet at the same point. The abc projective plane is usually called the **dual** of the xyz projective plane.

Proof. Suppose that we have a line in $\mathbb{P}_k^2$, which is given by $ax + by + cz = 0$ where a, b, c are not all 0, then we map it to the point $[a : b : c]$ in the abc plane. Notice that since the line equation is invariant under multiplication, this indeed gives a well defined function. This is clearly surjective to all the points in $\mathbb{P}_k^2$. The previous proposition shows that it's also injective. Clearly, we can construct a similar correspondence from the abc -plane to the xyz plane.

Now suppose that $p_1, \dots, p_n$ are $[x_1 : y_1 : z_1], \dots, [x_n : y_n : z_n]$ and they line on the same line:

$$Ax + By + Cz = 0$$

Then consider the lines in the abc plane:

$$x_i a + y_i b + z_i c = 0$$

We know that by our assumption, $x_i A + y_i B + z_i C = 0$, therefore these lines all intersect at the point $[A : B : C]$. The converse is clearly proved the same way. $\square$

Example 1.4. In this examples, let's study the usual **affine real conics** in the projective plane. Recall that in the affine x, y plane, a parabola, an ellipse and a hyperbola are given by the following equations:

$$F_1(x, y) = y - x^2 = 0, \quad F_2(x, y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} = 0, \quad F_3(x, y) = \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

Notice that the corresponding projective completion:

$$\tilde{F}_1(x, y, z) = yz - x^2 = 0, \quad \tilde{F}_2(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} = 0, \quad \tilde{F}_3(x, y, z) = \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

Recall that to compute the points at infinity, we set $z = 0$, and we get:

$$x^2 = 0, \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 0, \quad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

Clearly, for the first equation, we know that the only solution is $x = 0$, and we know that the only points at infinity on the parabola is $[0 : 1 : 0]$ and this matches our intuition.

We know that to get to the points on the line at infinity, we need to make the coordinates x, y approach infinity, which means that we are following the points of the parabola on the plane $z = 1$ and heading off toward the direction where x, y coordinates are growing. If we consider the line that connects the origin and the point on the parabola $(x, y, 1)$, then in the limit case, we approach the y -axis, which means that the point is $[0 : 1 : 0]$.

Similarly, we can do the computation for the ellipse and we need to solve:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$$

this gives us that $x = 0 = y$, which is not a point in the projective plane, therefore there is no points at infinity that lies on the ellipse.

Indeed, we know that if we follow the points on the ellipse and consider the line connecting it with the origin, we see that we never can approach those points at infinity.

Finally, we need to compute:

$$\frac{x^2}{a^2} = \frac{y^2}{b^2}$$

We know that $x = \pm \frac{a}{b}y$, this means that there will be two points at infinity that lies on the hyperbola, which are $[1 : \frac{a}{b} : 0]$ and $[1 : -\frac{a}{b} : 0]$.

If you follow the points on the hyperbola that goes to infinity and consider the line connecting them with the origin, we find the line gradually approaches the line on the xy -plane with either slope $\frac{a}{b}$ or $-\frac{a}{b}$, and these two lines exactly correspond to the two points $[1 : \frac{a}{b} : 0]$ and $[1 : -\frac{a}{b} : 0]$.

Example 1.5. In this example, let's carefully examine the relation between being smooth in the projective plane and being smooth in the affine plane. To proceed, let's consider the following useful result.

Let K be any field and let $\tilde{F}(x, y, z) \in K[x, y, z]$ be a homogeneous polynomial of degree n . Then we have the following relation:

$$x \frac{\partial \tilde{F}}{\partial x} + y \frac{\partial \tilde{F}}{\partial y} + z \frac{\partial \tilde{F}}{\partial z} = n \tilde{F}(x, y, z)$$

This is sometimes called **Euler's relation** and it's going to be very useful for our study of homogeneous polynomials. It's actually easy to prove it as we know that once you do the partial derivative of a monomial and multiply the variable back, you basically just have the monomial back but with a constant multiple which is the degree of that variable in that monomial. Notice that this relation has nothing to do with the characteristic of the field. But for our further argument, let's still assume that $\text{char}(K) = 0$.

Now suppose that we have a curve $\tilde{F}(x, y, z)$ defined by a homogeneous polynomial in the projective plane. We will define that the curve at $[x_0 : y_0 : z_0]$ is smooth iff the partial derivative vector doesn't vanish at (x_0, y_0, z_0) :

$$\begin{pmatrix} \frac{\partial \tilde{F}(x, y, z)}{\partial x} \big|_{(x_0, y_0, z_0)} \\ \frac{\partial \tilde{F}(x, y, z)}{\partial y} \big|_{(x_0, y_0, z_0)} \\ \frac{\partial \tilde{F}(x, y, z)}{\partial z} \big|_{(x_0, y_0, z_0)} \end{pmatrix}$$

This definition is natural because it directly generalizes the smoothness definition we have for an affine curve, not just they look similar, but with a deeper connection.

Suppose that $\tilde{F}$ is the homogenization of an affine plane curve $F(x, y)$, and the point of interest is $[x_0 : y_0 : z_0]$ with $z_0 \neq 0$. Then we know that we can think of $F(x, y)$ as $F(x, y, 1)$. I claim that using the definition of smoothness we gave above, $\tilde{F}(x, y, z)$ is smooth at $[x_0 : y_0 : z_0]$ iff $F(x, y)$ is smooth at $(\frac{x_0}{z_0}, \frac{y_0}{z_0})$. Indeed, if we assume that $\tilde{F}$ is smooth, then we know that the following vector is not 0:

$$\begin{pmatrix} \frac{\partial \tilde{F}(x, y, z)}{\partial x} \big|_{(x_0, y_0, z_0)} \\ \frac{\partial \tilde{F}(x, y, z)}{\partial y} \big|_{(x_0, y_0, z_0)} \\ \frac{\partial \tilde{F}(x, y, z)}{\partial z} \big|_{(x_0, y_0, z_0)} \end{pmatrix}$$

Suppose first that one of the first 2 coordinates is not 0, say the first coordinate is not 0. We know that the partial derivative is still homogeneous of degree $n - 1$ and since $z_0 \neq 0$, we can divide both side by z_0^{n-1} and I will let you show that this value is the same as:

$$\frac{\partial F(x, y)}{\partial x} \big|_{(\frac{x_0}{z_0}, \frac{y_0}{z_0})}$$

Since taking derivatives is additive, we can show this by showing this to every term of $F(x, y)$ and I am pretty sure everyone can do it.

If the first 2 coordinates are 0, then consider the relation:

$$x \frac{\partial \tilde{F}}{\partial x} \big|_{(x_0, y_0, z_0)} + y \frac{\partial \tilde{F}}{\partial y} \big|_{(x_0, y_0, z_0)} + z \frac{\partial \tilde{F}}{\partial z} \big|_{(x_0, y_0, z_0)} = n \tilde{F}(x, y, z) \big|_{(x_0, y_0, z_0)}$$

which implies that:

$$z \frac{\partial \tilde{F}}{\partial z} \big|_{(x_0, y_0, z_0)} = n \tilde{F}(x, y, z) \big|_{(x_0, y_0, z_0)} = 0$$

The last equation holds since we assume that $[x_0 : y_0 : z_0]$ is on the curve, which is a contradiction.

Of course, instead of assuming z_0 is not 0, we can assume x_0 or y_0 is not zero. But no matter what we assume, similar arguments as above will show that smoothness definition of $\tilde{F}$ we give implies F is smooth.

Conversely, I will also let you show that $F(x, y)$ is smooth at $(\frac{x_0}{z_0}, \frac{y_0}{z_0})$ implies $\tilde{F}$ is smooth at $[x_0 : y_0 : z_0]$, which is the easier direction. Therefore we know that the smoothness definition we make for $\tilde{F}$ indeed makes sense and is very natural.

Remark 1.6. Notice that in the previous example, we assume the point $[x : y : z]$ is already on the curve defined by $\tilde{F}$ then we showed that smoothness defined by $\tilde{F}$ and F are actually the same. Equivalently, the point $[x : y : z]$ is not smooth on the curve defined by $\tilde{F}$ iff $(\frac{x}{z}, \frac{y}{z})$ is not smooth on the curve F .

However, if we don't assume $[x : y : z]$ is on the curve, then if we want to use only the fact that the partial derivatives all vanish to conclude that $[x : y : z]$ is a nonsmooth point on the curve, we will have to require $\text{char}(K)$ doesn't divide n , or in particular $\text{char} K = 0$. To see why, notice that we will need to show that $\tilde{F}(x, y, z) = 0$, but we can only conclude $n\tilde{F}$ is 0, therefore we will need $n \neq 0$ to conclude $\tilde{F}$ is not 0.

Example 1.7. Recall that if we have a conic section in $\mathbb{P}_k^2$ defined by the following polynomial:

$$\tilde{F}[x, y, z] \in K[x, y, z], \quad \tilde{F}(x, y, z) = ax^2 + by^2 + cz^2 + dxy + exz + fyz$$

We will say that the conic section not degenerate if it's symmetric matrix:

$$A = \begin{pmatrix} a & \frac{d}{2} & \frac{e}{2} \\ \frac{d}{2} & b & \frac{f}{2} \\ \frac{e}{2} & \frac{f}{2} & c \end{pmatrix}$$

has nonzero determinant, i.e. $\det A \neq 0$.

The above definition is equivalent to say that the curve defined by $\tilde{F}$ in $\mathbb{P}_k^2$ is smooth at every point on the curve. To see why, let's first assume the matrix A is invertible. Then notice that the partial derivative vector:

$$\Delta \tilde{F}(x, y, z) = 2A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Then we know that since A is invertible, there is no nonzero $(x, y, z)^T$ such that $2A(x, y, z)^T = 0$, which means that $\tilde{F}$ is smooth at every point on the conic section. Conversely, if it's smooth at every point, we want to prove that A is invertible. Suppose that A is not invertible, then there is (x, y, z) such that:

$$A(x, y, z)^T = 0$$

Notice that this implies that:

$$(x, y, z)A(x, y, z)^T = 0$$

which means that (x, y, z) is on the conic section. But since $\Delta \tilde{F}(x, y, z) = 2A(x, y, z)^T = 0$, it contradicts to the hypothesis we assume that every point on the section is smooth.

Example 1.8. In this example, we go continue the previous example and show that any two nondegenerate conics over $\mathbb{P}_{\mathbb{R}}^2$ such that each of them at least have a real point in $\mathbb{P}_{\mathbb{R}}^2$ and are defined by matrices A, A' respectively are equivalent in the sense that via a linear transformation of coordinates we can get the conic section defined by A' using the conic section defined by A . The key here is to understand when can we **diagonalize** a matrix over a field K .

Recall that by **the Spectral Theorem**, if we work with a Hermitian matrix $A \in \text{GL}_n(\mathbb{C})$ (which simply means that $A^* = A$ where A^* is the complex conjugate transpose of A), then there will exists a unitary matrix U with $U^*U = \text{Id}_{n \times n}$ such that:

$$U^*AU = D$$

where D is an invertible diagonal matrix. If you can recall the proof of this result, then you will find that if we start with a real symmetric matrix A , then you will get a orthogonal real matrix P , instead of U and we have $P^TAP = D$ where D is a real matrix instead of a complex matrix.

This will suffice for to prove this, recall that if we are given a symmetric invertible real matrix A , then by what we just found, there is an orthogonal matrix P such that $P^TAP = D$ where D is diagonal and consists of all the eigenvalues of A . Since A is invertible, not eigenvalue can be 0, therefore we know that if we consider the linear transformation:

$$P : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad x \mapsto Px$$

We know that the conic $x^T Ax$ is mapped to $x^t P^T A P x = x^t D_A x = \lambda_1 x^2 + \lambda_2 y^2 + \lambda_3 z^2$ where λ_i are eigenvalues of A . In general, if we only assume A is symmetric, some λ_i could be 0. Of course some of them can be negative and other can be positive. Consider the following triple:

$$(p, n, z) \in \mathbb{Z}^3$$

where p stands for the number of positive eigenvalues A has, n stands for the number negative eigenvalues of A and z stands for the number of 0 eigenvalues of A . This triple is called the **inertia** of the real quadratic form A . Moreover, up to an invertible linear transformation, the inertia completely characterizes real quadratic forms. This result is usually called the **Sylvester's inertia law**. Notice that the order of x, y, z are not important since we can always use linear transformations to switch x, y, z .

In our case since we assumed that the two conic sections at least have a real value on $\mathbb{P}_{\mathbb{R}}^2$, then I will let you see that the inertia can not be $(3, 0, 0)$ or $(0, 3, 0)$. Therefore we only have two possibilities, $(2, 1, 0)$ or $(1, 2, 0)$. Since multiply the equation by -1 switches the number of positive and negative eigenvalues while leaving the solution sets of $\tilde{F}$ stable, we know that we can always make the inertia of such two conic the same. Say that it is $(2, 1, 0)$. Then consider the real linear transformation:

$$d : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad (x, y, z)^T \mapsto \left(\frac{x}{\sqrt{|\lambda_1|}}, \frac{y}{\sqrt{|\lambda_2|}}, \frac{z}{\sqrt{|\lambda_3|}} \right)$$

Then we know that after we use this transformation, the original conic section is transformed to:

$$x^2 + y^2 \pm z^2 = 0$$

Similarly, we can also do the same process and make the conic defined by A' looks like the one above, therefore we are done.

Remark 1.9. Note that the above argument won't work if we work with $\mathbb{Q}$ instead of with $\mathbb{R}$. One simple reason is that for example $\sqrt{2}$ is not in $\mathbb{Q}$, so we can not do the square root linear transformation.

However, we can still do the diagonalization process even if we work with an arbitrary field k with $\text{char}(k) \neq 2$. It's just that over $\mathbb{R}$ or $\mathbb{C}$, we can use orthogonal or unitary matrices P to do the linear transformation where $P^* = P^{-1}$. If we work with an arbitrary field we only can do the diagonalization using $P^T A P$ where P is not assumed to be $P^T = P^{-1}$. For one simple reason, for an arbitrary field, we don't have inner product, therefore the self-adjoint trick we did before won't work. But let's still prove that we can do the diagonalization using some P . If you recall this from linear algebra, this is called **diagonalization by congruence**. If $P^T = P$, it's called **diagonalization by similarity**.

To see how to do this diagonalization by congruence, suppose that A is a symmetric matrix over K , then we consider the linear form:

$$B : K^n \times K^n \rightarrow K, \quad (u, v) \mapsto u^T A v$$

Suppose that we have the standard basis $e_1, \dots, e_n$ of K^n , we can define the matrix of B with respect to this basis to be:

$$M_e = (B(e_i, e_j))_{ij}$$

Notice that $M_e = A$. I claim that there is at least a nonzero $v \in K^n$ such that $B(v, v) \neq 0$. If otherwise $B(v, v) = 0$ for all $v \in K^n$, then we know that:

$$B(u + v, u + v) = B(v, v) + B(u, u) + 2B(u, v) = 0$$

Then since $\text{char}(k) \neq 0$, this implies that $B(u, v) = 0$ for all u, v , which means that $A = 0$ since we know that $a_{ij} = B(e_i, e_j)$ where e_i, e_j are the standard basis of K^n .

Then suppose that v is such an element that $B(v, v) \neq 0$, then consider:

$$v^\perp = \{u \in K^n : B(v, u) = 0\}$$

Clearly, this is a subspace of K^n which satisfies $v^\perp \cap Kv = \{0\}$. Given any $x \in K^n$, if $B(v, x) = c$, then we know that:

$$B(v, x - \frac{c}{B(v, v)}v) = B(v, x) - c = 0$$

This implies that $K^n = v^\perp \oplus Kv$, then we know that by choosing a basis of $v^\perp$, we know that we get a new basis v_i of K^n such that the matrix of B is:

$$M_v = \begin{pmatrix} B(v, v) & 0 \\ 0 & A' \end{pmatrix}$$

Moreover, A' is also a symmetric matrix. Notice that, if we assume that the transition matrix from e_i to v_i is P , then I will let you prove the matrix of B with respect to this new basis is given by:

$$M_v = P^T M_e P = P^T A P$$

Then since A' is symmetric with lower dimension, we can use induction to assume that there is Q such that $Q^T A' Q = D$, then we know that:

$$\begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}^T M_v \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix} = \begin{pmatrix} B(v, v) & 0 \\ 0 & D \end{pmatrix}$$

Then if we denote $P' = \begin{pmatrix} 1 & 0 \\ 0 & Q \end{pmatrix}$ we know that:

$$(PP')^T A (PP') = D$$

If you remember how the Spectral Theorem is proved, you will find the above proof very familiar. Indeed, in the proof of the Spectral theorem, we again have a bilinear form from $\mathbb{C}^n \times \mathbb{C}^n$ to $\mathbb{C}$, which is induced by the inner product which has nothing to do with the matrix A . Here the bilinear form is given by the data A . The difference of the two forms explains why here we can not P orthogonal in the sense that $P^T P = \text{Id}$. However, we can make P a matrix, whose columns are orthogonal with respect to the bilinear form B . I will let you think a bit what is the difference here.

Example 1.10. We then go back to study the sets defined by a homogeneous polynomial $\tilde{F}(x, y, z) \in K[x, y, z]$ of degree n in the projective plane $\mathbb{P}_K^2$, where $\text{char}(K) = 0$. In this example, we will see how the tangent line in $\mathbb{P}_K^2$ works.

Suppose that $[x_0 : y_0 : z_0]$ is a smooth point on the curve. Let's first assume that $z_0 \neq 0$, then we will define the tangent line to the curve at $[x_0 : y_0 : z_0]$ to be the projective completion of tangent line of $F(x, y) = \tilde{F}(x, y, 1)$ at the point $(\frac{x_0}{z_0}, \frac{y_0}{z_0})$. Recall that the tangent line of $F(x, y)$ at this point is given by:

$$\frac{\partial F(x, y)}{\partial x} \Big|_{(\frac{x_0}{z_0}, \frac{y_0}{z_0})} (x - \frac{x_0}{z_0}) + \frac{\partial F(x, y)}{\partial y} \Big|_{(\frac{x_0}{z_0}, \frac{y_0}{z_0})} (y - \frac{y_0}{z_0}) = 0$$

I will let you see that we have the following relation:

$$\frac{\partial F(x, y)}{\partial x} \Big|_{(\frac{x_0}{z_0}, \frac{y_0}{z_0})} = \frac{1}{z_0^{n-1}} \frac{\partial \tilde{F}(x, y, z)}{\partial x} \Big|_{(x_0, y_0, z_0)}, \quad \frac{\partial F(x, y)}{\partial y} \Big|_{(\frac{x_0}{z_0}, \frac{y_0}{z_0})} = \frac{1}{z_0^{n-1}} \frac{\partial \tilde{F}(x, y, z)}{\partial y} \Big|_{(x_0, y_0, z_0)}$$

Then we know that we can rewrite the equation of the tangent line using:

$$\frac{1}{z_0^{n-1}} \frac{\partial \tilde{F}(x, y, z)}{\partial x} \Big|_{(x_0, y_0, z_0)} (x - \frac{x_0}{z_0}) + \frac{1}{z_0^{n-1}} \frac{\partial \tilde{F}(x, y, z)}{\partial y} \Big|_{(x_0, y_0, z_0)} (y - \frac{y_0}{z_0}) = 0$$

Then we multiply both sides with z_0^{n-1} , we get:

$$\frac{\partial \tilde{F}(x, y, z)}{\partial x} \Big|_{(x_0, y_0, z_0)} (x - \frac{x_0}{z_0}) + \frac{\partial \tilde{F}(x, y, z)}{\partial y} \Big|_{(x_0, y_0, z_0)} (y - \frac{y_0}{z_0}) = 0$$

Notice that using Euler's relation and the fact that $\tilde{F}(x_0, y_0, z_0) = 0$, we can rewrite the tangent line as:

$$\frac{\partial \tilde{F}(x, y, z)}{\partial x} \Big|_{(x_0, y_0, z_0)} x + \frac{\partial \tilde{F}(x, y, z)}{\partial y} \Big|_{(x_0, y_0, z_0)} y + \frac{\partial \tilde{F}(x, y, z)}{\partial z} \Big|_{(x_0, y_0, z_0)} z = 0$$

This tells us the completion is:

$$\frac{\partial \tilde{F}(x, y, z)}{\partial x} \Big|_{(x_0, y_0, z_0)} x + \frac{\partial \tilde{F}(x, y, z)}{\partial y} \Big|_{(x_0, y_0, z_0)} y + \frac{\partial \tilde{F}(x, y, z)}{\partial z} \Big|_{(x_0, y_0, z_0)} z = 0$$

Clearly, even though we assumed that z_0 is not 0 in $[x_0 : y_0 : z_0]$ and derived the formula for the tangent line using $z_0 \neq 0$. This actually doesn't depend on the coordinate in the sense that if x_0 is also not zero, we can do the above procedure using $x_0 \neq 0$ and get exactly the same formula. I will leave the details to you.

Example 1.11. Recall that if k is now an arbitrary field and A is a 3×3 invertible matrix. Then clearly, we get a bijective map:

$$A : \mathbb{P}_k^2 \rightarrow \mathbb{P}_k^2, \quad [x : y : z] \mapsto A[x_0 : y_0 : z_0]$$

Since A is linear, this is well defined. If we have a curve C on $\mathbb{P}_k^2$ which is defined by $\tilde{F}(x, y, z)$, then we can also consider another curve C' defined by $\tilde{F}'(x', y', z')$ where $(x', y', z')^T = A(x, y, z)^T$. Now I claim that if $P = [x_0 : y_0 : z_0]$ is a smooth point on C iff $P' = AP$ is going to be a smooth point on C' .

We know that:

$$\Delta_{(x, y, z)} \tilde{F}'(x', y', z') = A \Delta \tilde{F}(x, y, z)$$

Since A is invertible, this means that $\Delta \tilde{F}(x, y, z)$ is not 0 iff $\Delta_{(x, y, z)} \tilde{F}'(x', y', z')$ is not 0.

For example, if we consider the curve C defined by $y^2 - xz$, which is the projective completion of the usual parabola, we know that $[1 : 2 : 4]$ is a smooth point on the curve as we know that the partial derivatives are given by $(-4, 4, -1)$. Now consider the matrix:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

We know that after we apply A , the new curve C' is given by $x^2 - yz$, then by the above argument, we know that $[2 : 1 : 4]$ is again a smooth point on C' . What this example is trying to tell you is that in the projective plane, when talking about smoothness, which is a geometric notion, there is no intrinsic preferred coordinate that affect whether a point is smooth or not, x, y, z and y, x, z coordinate are exactly the same.

Using similar argument as above, we can also prove that the tangent line to a smooth point is also independent of coordinates.

Example 1.12. Again, we work with $\mathbb{P}_k^2$ and let $P_1 = [x_1 : y_1 : z_1] \in \mathbb{P}_k^2$ and $P_2 = [x_2 : y_2 : z_2]$ be two points in the projective plane. Consider the line passing through P_1 and P_2 , let's try to find a good description of the line denoted by $\overline{P_1 P_2}$. Recall that a line in $\mathbb{P}_k^2$ is given by $ax + by + cz = 0$. From the assumption, we know that:

$$ax_1 + by_1 + cz_1 = 0 = ax_2 + by_2 + cz_2$$

Therefore we know that (a, b, c) should be orthogonal to (x_1, y_1, z_1) and (x_2, y_2, z_2) . Therefore one good option is to take (a, b, c) to be the cross product of (x_1, y_1, z_1) and (x_2, y_2, z_2) if the cross product is not 0. Notice that this is independent of the nonzero orthogonal element we choose since we know that (x_1, y_1, z_1) and (x_2, y_2, z_2) are linearly independent, therefore the orthogonal complement is dimension 1.

Clearly, for $s, t \in K$, we know that the elements $s[x_1 : y_1 : z_1] + t[x_2 : y_2 : z_2]$ are on the line we just computed because (a, b, c) is orthogonal to (x_1, y_1, z_1) and (x_2, y_2, z_2) . This means that we can consider the following map:

$$\mathbb{P}_k^1 \rightarrow \mathbb{P}_k^2, \quad [s : t] \mapsto s[x_1 : y_1 : z_1] + t[x_2 : y_2 : z_2]$$

Clearly this is well defined and we know that the image lies on the line $\overline{P_1 P_2}$. I claim that this map is bijective. It's easy to show that this is injective using the fact that (x_1, y_1, z_1) and (x_2, y_2, z_2) are linearly independent. Suppose that $[x : y : z]$ is on the line, which mean that (x, y, z) is orthogonal to (a, b, c) . Notice that the orthogonal complement of (a, b, c) is two dimensional and the span of (x_1, y_1, z_1) and (x_2, y_2, z_2) is in the orthogonal complement, then they span the orthogonal complement. Therefore there are s, t such that $(x, y, z) = s(x_1, y_1, z_1) + t(x_2, y_2, z_2)$. Therefore we know that the line we just obtained can be parametrized by points in $\mathbb{P}_k^1$.

Now if we let $s = 1$ and let t vary, we find that the line we get those points:

$$[x_1 + tx_2 : y_1 + ty_2 : z_1 + ty_2]$$

Notice that the is the image of the map $\mathbb{P}_k^1 \rightarrow \mathbb{P}_k^2$ except at the point $[0 : 1]$, notice that when we plug in $[0 : 1]$, we get P_2 , therefore we get every point on the line except the point P_2 .

Remark 1.13. Another geometric way to think about this line $\overline{P_1 P_2}$ is to think about the plane spanned by P_1, P_2 in $\mathbb{R}^3$, which consists of all points $sP_1 + tP_2$, then we do the projectivization, which means to think of the image of the points in $\mathbb{P}_k^2$. You can check the line you get is the above line you obtained using the algebraic method.

Now we want to discuss **the point of inflection**. Let's work in the xy affine plane and consider a curve $F(x, y) = 0$ defined by some polynomial in x, y . For simplicity, let's first assume that we are working with the field $\mathbb{R}$ or $\mathbb{C}$. Suppose that $P_1 = (x_1, y_1)$ is a smooth point of $F(x, y)$ where the tangent line is not vertical. Then recall that using the **Implicit Function Theorem**, we can locally write $y = f(x)$ for a smooth function f such that $f(x_1) = y_1$. If we do the Taylor expansion of f at x_1 , we know that the coefficient of the linear term is going to be the slope of the tangent line. We know that if we subtract off the linear term, we get:

$$f(x) - y_1 - f'(x_1)(x - x_1) = a_m(x - x_1)^m + \cdots, \quad a_m \neq 0$$

Here we will call this $m \geq 2$ **the order of tangency**. We will say that P_1 is a **point of inflection** if $m > 2$, i.e. $f''(x_1) = 0$. Now we assume that $F(x, y)$ comes from $\tilde{F}(x, y, 1)$ where $\tilde{F}(x, y, z)$ is a homogeneous polynomial. Moreover, if $P_1 = [x_1 : y_1 : z_1]$ with $z_1 \neq 0$ is a smooth point on the curve C defined by $\tilde{F}$ and we consider the tangent line ℓ of C at P_1 . If we pick another point $P_2 = [x_2 : y_2 : z_2]$ on ℓ that is different from P_1 , I claim that the order of tangency is the lower power of t showing up in $\tilde{F}(x_1 + tx_2, y_1 + ty_2, z_1 + tz_2)$.

Indeed, recall that the tangent line of C at P_1 is given by the following equation:

$$\frac{\tilde{F}(x, y, z)}{\partial x} \Big|_{(x_1, y_1, z_1)} x + \frac{\tilde{F}(x, y, z)}{\partial y} \Big|_{(x_1, y_1, z_1)} y + \frac{\tilde{F}(x, y, z)}{\partial z} \Big|_{(x_1, y_1, z_1)} z = 0$$

Then suppose that $P_2 = [x_2, y_2, z_2]$ is on this line, then and consider the function:

$$\tilde{F}(x_1 + tx_2, y_1 + ty_2, z_1 + tz_2) : K \rightarrow K$$

We know that $\tilde{F}(0) = 0$ and suppose that m' is the lowest degree of t , then m' is the first derivative we get such that when we evaluate at 0, it's not 0. To see this, consider $F(x, y) = \tilde{F}(x, y, 1)$. Let's assume that the y partial of $F(x, y)$ at the point $(\frac{x_1}{z_1}, \frac{y_1}{z_1})$ is not 0. For simplicity, let's denote this point by (x'_1, y'_1) . We will also see that you can also assume that the x partial is not 0 and it doesn't affect our later argument. Since the y partial is not 0, we can write the equation $F(x, y) = 0$ locally as $y = f(x)$ for a smooth function f and we know that:

$$f(x) - f(x - 1') - f'(x'_1)(x - x'_1) = a_m(x - x'_1)^m, \quad a_m \neq 0$$

Let's parametrize the tangent line at $(x'_1, f(x'_1))$ by $(x'_1 + t, f(x'_1) + f'(x'_1)t)$. Then **** to complete later* Notice that the above discussion actually has nothing to do with which coordinate is not 0, and it has nothing to do whether the tangent line is vertical or horizontal. I will let you think through it.

Proposition 1.14. The m defined above does not change under any change of coordinates, therefore we will define a point P on C to be an inflection point if $m > 2$. Therefore we know that after any change of coordinate, an inflection point still remains an inflection point. In particular, if $z = 0$, we can first switch the coordinate and use our familiar xy coordinate to do computations.

Proof. **** to complete later* □

Example 1.15. Suppose that $y^2 = f(x)$ is a real elliptic curve. We already know that $[0 : 1 : 0]$ is a point on the projective completion of this curve. I claim that $z = 0$, which is the line at infinity is tangent to the curve at $[0 : 1 : 0]$ and this point is a point of inflection. Indeed the projective completion is:

$$y^2 z = x^3 + ax^2 z + bxz^2 + cz^3$$

Then we know that the partials are:

$$(3x^2 + 2axz + bz^2, 2yz, y^2 + 2bxz + 3cz^2 + ax^2)$$

Then if we evaluate it at $[0 : 1 : 0]$, we get the line $z = 0$, therefore we know that $z = 0$ is indeed the tangent line. To see that $[0 : 1 : 0]$ is indeed a point of inflection, notice that we can take a point $[1 : 0 : 0]$ on the line $z = 0$. Then we consider:

$$\tilde{F}(t, 1, 0) = t^3$$

Therefore we know that $m = 3 > 2$, therefore it's indeed an inflection point.

1.4 Doubly Periodic Functions

Let L be a lattice in the complex plane which means that:

$$L = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$$

where ω_1, ω_2 are two $\mathbb{R}$ -linearly independent complex numbers. For example we can take $\omega_1 = i$ and $\omega_2 = 1$, then L will consist of all the **Gaussian integers**. It turns out that this lattice of Gaussian integers has a lot to do the elliptic curve $y^2 = x^3 - n^2x$ that comes from the congruence problem.

We define the **fundamental parallelogram** for ω_1, ω_2 by:

$$\Pi = \{a\omega_1 + b\omega_2 : a, b \in [0, 1]\}$$

In general we will order ω_1, ω_2 in clockwise order, which means that $\frac{\omega_1}{\omega_2}$ has positive imaginary part.

Since ω_1, ω_2 forms an $\mathbb{R}$ -basis of $\mathbb{C}$, we know that for any $x \in \mathbb{C}$, it has a unique representation as $x = a\omega_1 + b\omega_2$. If a and b are not integers, we know that we can write then uniquely as the sum of some element in L and some element in Π .

What we need to notice that to obtain the same lattice, we can have multiple choice of ω_1 and ω_2 . For example, ω_1, ω_2 and $\omega_1 + \omega_2, \omega_2$ gives you the same lattice.

Proposition 1.16. The lattice $L = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and the lattice $L' = \mathbb{Z}\omega'_1 + \mathbb{Z}\omega'_2$ are the same if there is some matrix $A \in \text{GL}_2(\mathbb{Z})$ such that $\omega' = A\omega$ where $\omega = (\omega_1, \omega_2)^T$ and similarly for ω' . If we insist on our convention that ω_1, ω_2 are clockwise, then A is actually in $\text{SL}_2(\mathbb{Z})$.

Proof. Clearly, if we denote the inverse of A to be B , we know that B is also a matrix with integer values and $AB = \text{Id} = BA$. Since the two determinants are all integers, then they have to be ± 1 . Then if we have $n\omega'_1 + m\omega'_2 = (n, m)\omega'$, we know that we can consider $(n, m)A\omega = (n, m)\omega'$. Conversely, if we have $n\omega_1 + m\omega_2$, we can use $(n, m)B\omega'$.

Conversely, consider the two points ω_1, ω_2 in $\mathbb{C}$, since the two lattice are the same, there will be $a\omega'_1 + b\omega'_2 = \omega_1$ and $c\omega'_1 + d\omega'_2 = \omega_2$. Then consider the matrix:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

We know that $A\omega' = \omega$. Similarly, we have B such that $B\omega = \omega'$, then we know that $AB\omega = \omega$, then since ω_1, ω_2 are $\mathbb{R}$ -linearly independent, we know that $AB = \text{Id}$, similarly $BA = \text{Id}$.

Now if we want $\omega' = A\omega$ and we want ω'_1, ω'_2 to be clockwise. We know that A has to preserve orientation, therefore we know that it must have determinant 1. $\square$

For a given lattice L , a meromorphic function $f(z)$ on $\mathbb{C}$ is said to be an **elliptic function relative to** L if $f(z + l) = f(z)$ for all $l \in L$. Notice that it suffice to check this property for $l = \omega_1$ and $l = \omega_2$. In other words, an elliptic function is just a periodic function with two periods. Such a function is completely determined by it's values on the fundamental parallelogram. For a given lattice L , we will use $\mathcal{E}_L$ to denote all such functions. Recall that the set of all meromorphic functions over $\mathbb{C}$ is a field, then it's clear that $\mathcal{E}_L$ is a subfield. Moreover, you can check that $\mathcal{E}_L$ is closed under differentiation.

Now we will give a sequence of proposition related to properties special to elliptic functions relative to L . It turns out that being an elliptic function is more restrictive than you probably would think.

Proposition 1.17. Let $f(z) \in \mathcal{E}_L$ such that it has no poles in Γ . Then $f(z)$ has to be constant over $\mathbb{C}$.

Proof. Recall that if f has no poles in Π , then it has no poles over $\mathbb{C}$, therefore it's indeed a holomorphic function. Since it's doubly periodic, it's values are determined by it's values in Π , therefore it's bounded, therefore we know that by Liouville's theorem, f is constant. $\square$

Notice that given a meromorphic function f and a bounded region, it at most has finitely many poles inside the closure of this region. This implies that if $f \in \mathcal{E}_L$, it at most have finitely many poles in Π , then I will let you show that there is an $\alpha \in \mathbb{C}$ such that the translate of Π , i.e. $\alpha + L$ contains no poles on the boundary C of $\alpha + \Pi$. Then we can apply the residue theorem:

Proposition 1.18. Using the same notation as above, then the sum of residues of $f(z)$ in Π is 0.

Proof. Notice that if we compute the integral:

$$\frac{1}{2\pi i} \int_C f(z) dz$$

By residue theorem, it's the sum of all the residues in $\Pi + \alpha$. However, since we have opposite directions for opposite sides but the same value on both sides, we know that the sum of integrals on opposite sides are 0, therefore the above integral is 0. $\square$

The above proposition implies that if $f \in \mathcal{E}_L$ is an elliptic function which is not constant, then it has poles in Π , since it only has finitely many poles in Π . If we assume that there is no poles along the boundary, then there are at least 2 poles in the interior since if there is only one, the residue summation is not 0. If there are indeed poles along the boundary, by slightly translating Π , we can assume that the poles all come from the boundary of Π and the interior of Π , and there are still at least 2 poles by the same reason.

Proposition 1.19. Using the above notation and suppose that $f(z)$ has no zeros and poles along the boundary of $\Pi + \alpha$, then if we let $\{m_i\}$ to be the order of the various zeros in the interior of $\alpha + \Pi$ and let $\{n_j\}$ be the orders of the various poles in the interior of $\Pi + \alpha$, then we have:

$$\sum_i m_i = \sum_j n_j$$

Proof. Consider the function $f'(z)$, which is still a elliptic function. We know that if we assume that f has no zeros or poles along the boundary, then $f'(z)$ has no poles along the boundary. Then consider the function:

$$\frac{f'(z)}{f(z)}$$

which is again an elliptic function relative to L which has no poles on the boundary of $\alpha + \Pi$. Then the sum of the residue of $\frac{f'(z)}{f(z)}$ is on the first hand 0 by the previous proposition. It's also the summation of the order of the zeros and poles of f , therefore we are done. $\square$

Now we are going to define an example which will turn out to be a key example of elliptic functions relative to the lattice $L = \{m\omega_1 + n\omega_2\}$. The function is called the **Weierstrass $\wp$ function**. Usually, it's denoted by $\wp(z; L)$ or $\wp(z; \omega_1, \omega_2)$. If the lattice is clear, we sometimes just write $\wp(z)$. We will set:

$$\wp(z; L) = \frac{1}{z^2} + \sum_{l \in L, l \neq 0} \left(\frac{1}{(z-l)^2} - \frac{1}{l^2} \right)$$

Proposition 1.20. The sum defined above converges absolutely and uniformly for all z in any compact subset of $\mathbb{C} \setminus L$.

To prove the convergence, recall that if we work with the 1-dimensional real number case and have a sum $\frac{1}{x} + \sum \left(\frac{1}{x-l} + \frac{1}{l} \right)$ where l is taken over nonzero $l \in \mathbb{Z}$. We would like to prove that this sum converges absolutely and uniformly on any compact subset of $\mathbb{R} \setminus \mathbb{Z}$. After you see how this one dimensional case is proved you will understand how to prove the complex number case.

Notice that the summand can be rewritten as $\frac{x}{l(x-l)}$. We would like to use a comparison set which is $\frac{1}{l^2}$ and use that this summation on comparison set converges to show that the above summation converges absolutely and uniformly. More specifically, if we have a convergent sequence $\sum_l b_l$ of positive real numbers and $f_l(x)$ satisfies the property that $|\frac{f_l(x)}{b_l}|$ approaches a finite limit as $l \rightarrow \pm\infty$ uniformly for x in some set K . Then $\sum_l f_l(x)$ converges absolutely and uniformly over that set K . I will let you fill in the details.

The proof for the convergence of the complex series is done similarly, except that $\mathbb{Z}$ is now replaced with L . We know that:

$$\frac{1}{(z-l)^2} - \frac{1}{l^2} = \frac{2lz - z^2}{l^2(z-l)^2}$$

Then for $z \in K$ where K is a compact subset of $\mathbb{C} \setminus L$, we know that we can assume that $|z| \leq R$, then we know that if we consider:

$$\frac{l^3(2lz - z^2)}{l^2(z - l)^2} = \frac{l(2lz - z^2)}{(z - l)^2}$$

Notice that if we take absolute value, we have:

$$\left| \frac{l(2z - z^2)}{(z - l)^2} \right| = \left| \frac{(2z - \frac{z}{l}z)}{(\frac{z}{l} - 1)^2} \right|$$

If we now let $l \rightarrow \pm\infty$, then you can see that this limit uniformly goes to $2z$ and since z is in K on a compact set we know that the limit is finite and has norm at most R for some finite R . Then we know that we can conclude from $\sum_{l \in L} \frac{1}{|l|^3}$ converges that the summation converges absolutely and uniformly on K . Finally, the fact that $\sum \frac{1}{|l|^3}$ converges comes from the fact that:

Lemma 1.21. $\sum_{l \in L} |l|^{-s}$ converges if $s > 2$.

Proof. Notice that we can group the summands by considering those l which satisfies $n - 1 < |l| \leq n$ to be in the same group. I claim that there is a constant $C > 0$ such that the number of such l in the annulus is bounded by Cn . To see why this holds, we can assume that $|\omega_1|$ and $|\omega_2|$ are both 1 since this only will let our bound differ by a constant, then consider the line spanned by ω_1 , we know that on this line, there are only 2 points in the annulus. Then we consider the line $\omega_2 + t\omega_1$, then this line has at most 2 points in the annulus, then we consider the line $-\omega_2 + t\omega_1$, again at most two points. Notice that we need to do this process $\frac{n}{|x_2|}$ where x_2 is the x coordinate of ω_2 , this gives you a constant times n .

Therefore, this summation is bounded by:

$$\sum_n n \frac{1}{n^s} = \sum n^{1-s}$$

We know that this series converges when $s - 1 > 1$ which means $s > 2$. □

Proposition 1.22. $\wp(z; L)$ is in $\mathcal{E}_L$ and it's only poles are all double pole at each lattice point.

Proof. Clearly, when we consider any compact subset of $\mathbb{C} \setminus L$, we know that the summation that defines $\wp(z; L)$ converges uniformly. Notice that each of the summands is continuous on this compact subset, therefore a uniform limit of continuous functions is still continuous. This tells you that points outside of L are not poles of $\wp(z; L)$. Moreover, for any $z \in \mathbb{C} \setminus L$, we can pick a open disk around z contained in $\mathbb{C} \setminus L$ and conclude that $\wp(z; L)$ is actually holomorphic around z since it's a uniform limit of holomorphic functions.

For $l \in L$, consider the function:

$$\wp(z; L) - \frac{1}{(z - l)^2}$$

We know that we can take a closed disk around l and conclude that $\wp(z; L) - \frac{1}{(z - l)^2}$ is actually holomorphic in a neighborhood of l . Since $\frac{1}{(z - l)^2}$ has a pole of order 2 at l , then so is $\wp(z; L)$. This tells you that $\wp$ is actually a meromorphic function on $\mathbb{C}$ with a double pole at each lattice point.

To prove it's doubly periodic, we first notice that $\wp$ is actually an even function in the sense that $\wp(z) = \wp(-z)$. Indeed, we know that $\wp(-z)$, by definition is the same sum with that of $\wp(z)$. Then we consider the derivative of $\wp(z)$ at neighborhood of points away from points in L . Since it's the uniform limit of holomorphic functions, we can take derivative term by term and consider the limit, which means that:

$$\wp'(z; L) = -2 \sum_{l \in L} \frac{1}{(z - l)^3}$$

Clearly $\wp'(z; L)$ is doubly periodic with respect to L . Then to prove $\wp(z; L)$ is doubly periodic with respect to L , we need to prove that for $z \in \mathbb{C} \setminus L$ and ω_1, ω_2 , we have:

$$\wp(z + \omega_i) - \wp(z) = 0$$

Notice that the derivative of the function $\wp(z + \omega_i) - \wp(z)$ is $\wp'(z + \omega_i) - \wp'(z) = 0$ since we already know that $\wp'$ is doubly periodic. Then we must have $\wp(z + \omega_i) - \wp(z)$ is some constant C for all $z \in \mathbb{C} \setminus L$. Now let $z = -\frac{1}{2}\omega_i$, we know that:

$$\wp\left(\frac{1}{2}\omega_i\right) - \wp\left(-\frac{1}{2}\omega_i\right) = C$$

However, recall that $\wp$ is even, therefore:

$$C = \wp\left(\frac{1}{2}\omega_i\right) - \wp\left(-\frac{1}{2}\omega_i\right) = \wp\left(\frac{1}{2}\omega_i\right) - \wp\left(\frac{1}{2}\omega_i\right) = 0$$

□

Notice that the fact that $\wp(z; L)$ is doubly periodic is not clear from the definition of it.

Remark 1.23. After we know that the only poles of $\wp$ are on the lattice. We know that we can pick a small α such that $\wp$ has no poles and zeros along $\Pi + \alpha$. Since the order of poles is 2 inside $\alpha + \Pi$, we know that the order of zero is also 2 in $\Pi + \alpha$. It could be a double 0 or two simple zeros. We can conclude the same for $\wp(z; L) - u$ for any constant u and therefore we know that $\wp(z)$ assumes the u in $\Pi + \alpha$, either once by with multiplicity 2 or twice at two different points.

Again, since $\wp$ is doubly periodic, we can also assume that in Π , $\wp$ also assume every point $\mathbb{C} \cup \{\infty\}$ twice with appropriate multiplicity. We will give more details in the exercises of next section.

1.5 The Field of Elliptic Functions

In the previous section, we gave a concrete example of an elliptic function associated to a lattice L . It turns out that, just like how $\sin$ and $\cos$ play a key role for periodic real valued functions via Fourier series, the Weierstrass $\wp(z)$ function and it's derivative $\wp'(z)$ play a keep role for all elliptic functions. But unlike the real valued function, where we need to use infinite series, for elliptic functions, it's simpler to express every elliptic function in terms of the two basic ones.

Proposition 1.24. $\mathcal{E}_L = \mathbb{C}(\wp(z; L), \wp'(z; L))$. That is to say that any elliptic function relative to the lattice L can be written as a rational function of $\wp(z; L)$ and $\wp'(z; L)$. More precisely, given $f(z) \in \mathcal{E}_L$, there are two rational functions $g(X), h(Y)$ with coefficients in $\mathbb{C}$ such that $f(z) = g(\wp(z; L)) + \wp'(z; L)h(\wp(z; L))$.

Proof. Note that if $f(z) \in \mathcal{E}_L$, then so are the two even functions:

$$\frac{f(z) + f(-z)}{2}, \quad \frac{f(z) - f(-z)}{2\wp'(z)}$$

It's clear that the first one is an even function. Notice that by definition:

$$\wp'(z) = -2 \sum_{l \in L} \frac{1}{(z - l)^3}$$

This implies that $\wp'(-z) = -\wp'(z)$, then clearly the second one is also an even function.

Notice that we have:

$$f(z) = \frac{f(z) + f(-z)}{2} + \wp'(z) \frac{f(z) - f(-z)}{2\wp'(z)}$$

□

Then we know that to prove the above proposition, it will suffice to show that:

Proposition 1.25. The subfield $\mathcal{E}_L^+ \subset \mathcal{E}_L$ of even elliptic functions relative to L is $\mathbb{C}(\wp(z))$. In other words, given any such function f , there is a rational function $R_1(z)$ such that $f(z) = R_1(\wp(z))$.

I will refer you to go check out the proof in the note series of algebraic geometry on compact Riemann surfaces as the proof gives you a better understanding of $\wp'(z), \wp(z)$. But what is important in this proof is that this proof is constructional and given an even elliptic function f , we can cook up a rational function $R_1(z)$ so that $f(z) = R_1(\wp(z))$. Once you understand the proof given there, without more work, we know that:

Corollary 1.26. $\wp'(z)^2$ is a cubic polynomial in $\wp(z)$.

Proof. Recall that the whole point of the proof of the above proposition is to find a rational expression of $\wp'(z)^2$ and a rational expression of $\wp(z)$ such that they have the same "easy-to-understand" zeros and poles. Then they only differ by a scalar multiple. For a general even elliptic function f , we don't know much of it's poles, therefore we choose to don't care it's poles by dividing something with the same pole, which gives you something only with simple zeros and poles. However, we know a lot about $\wp'(z)^2$ to save us from all of the work.

Here we know that if we choose some $c \in \mathbb{C}$ such that the zeros of $\wp'(z)^2 - c$ are all simple zeros $\{a_1, -a_1, a_2, -a_2, a_3, -a_3\}$ such that they are not half lattice points. Clearly the only poles are at lattice points and are of order 6. We also know that the only poles are at lattice points. Now consider the product:

$$(\wp(z) - \wp(a_1))(\wp(z) - \wp(a_2))(\wp(z) - \wp(a_3))$$

We know that it has simple zeros at $\{a_1, -a_1, a_2, -a_2, a_3, -a_3\}$ and degree 6 pole at lattice points. Therefore we know that:

$$\wp'(z)^2 - c = \lambda((\wp(z) - \wp(a_1))(\wp(z) - \wp(a_2))(\wp(z) - \wp(a_3)))$$

This implies the conclusion. $\square$

Corollary 1.27. The even function $\wp(Nz)$ is also a rational expression of $\wp(z)$.

Both of these facts turns out to be more important as we dive deeper into the study. The first tells us that the Weierstrass polynomial satisfies an equation of a very special type which gives connections between elliptic curves. The second fact is the starting point for studying **points of finite order** on elliptic curves. We will give the details as we go down.

Below are the exercises in this exercise:

Proposition 1.28. There is a continuous group isomorphism from $\mathbb{C}/L$ to $\mathbb{S}^1 \times \mathbb{S}^1$ where $\mathbb{S}^1$ is the circle group in the complex plane. However, in general this is not a biholomorphism.

Proof. Recall that we have a natural Riemann surface structure on $\mathbb{S}^1 \times \mathbb{S}^1$ where $\mathbb{S}^1$ if we locally think of the points on $\mathbb{S}^1 \times \mathbb{S}^1$ where $\mathbb{S}^1$ as angles. I will refer you to go to the Riemann surface note series to review why $\mathbb{C}/L$ is a Riemann surface. Clearly they both are compact.

Consider the map:

$$f : \mathbb{C} \rightarrow \mathbb{S}^1 \times \mathbb{S}^1, \quad (a\omega_1 + b\omega_2) \mapsto (e^{a2\pi i}, e^{b2\pi i})$$

Clearly this map is in general not holomorphic since basically you are just trying to prove that the following map is holomorphic:

$$z = x + iy \mapsto a(z) + c + i(b(z) + d)$$

Then if you compute the derivative of this map, it's:

$$\begin{pmatrix} \omega_1 & \omega_2 \end{pmatrix}^{-1}$$

Therefore unless $\begin{pmatrix} \omega_1 & \omega_2 \end{pmatrix}^{-1}$ this matrix represents a complex number, this map is not holomorphic.

However, this map is indeed continuous and notice that clearly that this is constant on the fibers $\mathbb{C} \rightarrow \mathbb{C}/L$, therefore it uniquely factors through $\mathbb{C}/L$ and we get the following diagram:

$$\begin{array}{ccc} \mathbb{C} & \xrightarrow{f} & \mathbb{S}^1 \times \mathbb{S}^1 \\ & \searrow \pi & \nearrow \\ & \mathbb{C}/L & \end{array}$$

Notice that this is obviously bijective and it's from a compact space to a Hausdorff space, therefore it's a homeomorphism.

This is even a homeomorphism of topological groups since clearly it respects addition. $\square$

Using this isomorphism of topological groups, we can compute:

Problem 1.29. How many points of order N or a divisor of N are in the group $\mathbb{C}/L$.

Solution. We can just think about how many elements of order N in $\mathbb{S}^1 \times \mathbb{S}^1$.

Recall that if we ask how many elements of order N is in $\mathbb{S}^1$, the number is just the number of primitive N -th roots of unity and we know the number is $\varphi(N)$ where φ is the Euler's totient function. If we want to compute the elements of order N in $\mathbb{S}^1 \times \mathbb{S}^1$, we need to use the **second Jordan totient function**:

$$J_2(N) = N^2 \prod_{p|N} \left(1 - \frac{1}{p^2}\right)$$

To see why, suppose that we have some element whose order divide N :

$$\left(e^{\frac{2\pi ai}{N}}, e^{\frac{2\pi bi}{N}}\right)$$

Then we know that its order is:

$$\frac{N}{\gcd\{a, b, N\}}$$

Therefore, if the order is exactly n , $\gcd\{a, b, N\} = 1$, therefore we need to find those pairs in $\mathbb{Z}/N \times \mathbb{Z}/N$ such that one of the two coordinates should be relatively prime to N , which means both the coordinates can not have a common prime with N . Notice that given a prime $p \mid N$, the portion of pairs whose both coordinates have a factor p is $\frac{1}{p^2}$, therefore the rest $1 - \frac{1}{p^2}$ are the ones we can choose. This gives the above function.

If we want to find elements whose order divides N , then it's just $\mu_N \times \mu_N \simeq (\mathbb{Z}/N)^2$.

Problem 1.30. Show that the subgroups of $\mathbb{C}/L$ of prime order p is in one to one correspondence with points in $\mathbb{P}_{\mathbb{F}_p}^1$ and count how many of them.

Solution. Recall that $\mathbb{P}_{\mathbb{F}_p}^1$ only has $p + 1$ elements. Let's think about how many subgroup of order p there are in $\mathbb{S}^1 \times \mathbb{S}^1$. Notice that if G is such a group, then we pick any nonzero element in this group and consider the subgroup it generates, this turns out to be the whole group. Therefore we know that every such group contains p elements, $p - 1$ of them are all order p elements. Notice that the number of order p elements in $\mathbb{S}^1 \times \mathbb{S}^1$ is:

$$p^2 \left(1 - \frac{1}{p^2}\right) = p^2 - 1 = (p + 1)(p - 1)$$

Then we know that this will correspond to $p + 1$ such subgroups. Notice that these elements are all in:

$$\mathbb{Z}/p \times \mathbb{Z}/p$$

Then we know that except $(0, 0)$, each of them is an element with order p and we can define a map:

$$(\mathbb{Z}/p \times \mathbb{Z}/p) \setminus \{(0, 0)\} \rightarrow \mathbb{P}_{\mathbb{F}_p}^1, \quad (a, b) \mapsto [a : b]$$

Clearly this is surjective. Notice that if (a, b) and (c, d) are in the same subgroup, since the subgroup is of order p , we know that (a, b) is a generator, therefore $(c, d) = (a, b)^i$ therefore we know that (c, d) and (a, b) are mapped to the same element in $\mathbb{P}_{\mathbb{F}_p}^1$, therefore the above map descends to a map from the order p -subgroups to $\mathbb{P}_{\mathbb{F}_p}^1$ and it's again surjective. To show it's injective if two groups are mapped to the same element, we know that the image is a generator of the group, therefore the two groups are equal.

Problem 1.31. Let $s = 2, 3, 4, \dots$. Fix a positive integer N , and let $f : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{C}$ be any function of period N , that is:

$$f(m + N, n) = f(m, n) = f(m, n + N)$$

Suppose that $f(0, 0) = 0$ and if $s = 2$ further suppose that $\sum f(m, n) = 0$ where the summation is over $0 \leq m, n < N$. Now we define a function:

$$F_s(\omega_1, \omega_2) = \sum_{m, n \in \mathbb{Z}} \frac{f(m, n)}{(m\omega_1 + n\omega_2)^s}$$

Then prove that this series converge absolutely when $s > 2$ and converges conditionally when $s = 2$. of course, when we say something converge conditionally, we have to specify a order of summation. We will take m, n in nondecreasing order of $|m\omega_1 + n\omega_2|$.

Solution. When $s > 2$, consider:

$$\sum_{m,n \in \mathbb{Z}} \left| \frac{f(m,n)}{(m\omega_1 + n\omega_2)^s} \right|$$

Let's assume that the maximal moduli of $f(m,n)$ is M , then we know that we have:

$$\sum_{m,n \in \mathbb{Z}} \left| \frac{f(m,n)}{(m\omega_1 + n\omega_2)^s} \right| \leq \sum_{m,n \in \mathbb{Z}} \left| \frac{M}{(m\omega_1 + n\omega_2)^s} \right| \leq \sum_{\omega \in \Omega^*} \frac{M}{|\omega|^s}$$

I will let you prove that such a series converges iff $s > 2$. Therefore when $s > 2$, the original series converges absolutely.

When $s = 0$, we want to prove that in the order we specified before, the double series converges. Let's assume that function $f(m,n)$ is not 0 for all m,n , otherwise there is nothing to prove. To prove that this converges, let's prove first that using another order of summation, the series converges. Let's first partition $\mathbb{Z} \times \mathbb{Z}$ into $N \times N$ blocks, that is we want to write each element in the product as:

$$(aN + r, bN + s), \quad 0 \leq r, s < N$$

Set:

$$q_{ab} = N(a\omega_1 + b\omega_2), \quad \delta_{r,s} = r\omega_1 + s\omega_2$$

We know that by periodicity:

$$f(aN + r, bN + s) = f(r, s)$$

Then we consider the sum over the block a, b , which is:

$$B_{a,b} = \sum_{r,s} \frac{f(r,s)}{(q_{a,b} + \delta_{r,s})^2}$$

Notice that we can use $\sum_{r,s} f(r,s) = 0$ to conclude that:

$$B_{a,b} = \sum_{r,s} \frac{f(r,s)}{(q_{a,b} + \delta_{r,s})^2} = \sum_{r,s} f(r,s) \left(\frac{1}{(q_{a,b} + \delta_{r,s})^2} - \frac{1}{q_{a,b}^2} \right) = \sum_{r,s} f(r,s) \frac{2\delta_{r,s} + \frac{\delta_{r,s}^2}{q_{a,b}}}{q_{a,b}(q_{a,b} + \delta_{r,s})^2}$$

Then we consider:

$$\sum_{a,b \in \mathbb{Z}} B_{a,b}$$

I will let you prove this converges absolutely and further more, the sum converges absolutely using this way to group them also implies that the original summation converges.

However, the above order is not the order we specified where we want to sum in the order where $|m\omega_1 + n\omega_2|$ is non decreasing. A useful conceptual tool is to think about the sum with respect to the radius. Let $\lambda_{m,n} = m\omega_1 + n\omega_2$, then for each $R > 0$, we define:

$$S(R) = \sum_{0 < |\lambda_{m,n}| \leq R} \frac{f(m,n)}{\lambda_{m,n}^2}$$

Let's compare this with the summation over the $N \times N$ block. Let $q_{a,b}$ be a point lying inside the disk of radius R . Let $D = \max_{0 \leq r,s < N} |\delta_{r,s}|$, therefore every point in the block lies within distance D to $q_{a,b}$, therefore the block is only partially contained in R iff $|q_{a,b}| + D > R$ or $|q_{a,b}| - D < -R$, which means that:

$$R - D < |q_{a,b}| < D - R$$

Notice that this means that when $q_{a,b}$ is in the annulus with width $2D$, the block is contained only partially in the disk. However, notice that $2D$ is fixed, therefore as R grows larger, we know that the width is going to be insignificant compared to R , therefore we know that those bad blocks (which are those not completely contained in the disk) at most have N^2 times CR points for some constant C . Moreover, we know that the blocks on the boundary has magnitude at most $\frac{C}{R}$, therefore we know that as R grows larger, the difference between the block sum and the radius sum is going to be $\frac{CN^2R}{R^2} = \frac{CN^2}{R} \rightarrow 0$, therefore we know that after we regroup the block sums (which is fine since it's absolutely convergent), the limit of the radius sum is the same as the block sum. Notice that we notice that each of the term in the prescribed sub only differs with some possible later terms in the radius sum by $\frac{C}{R^2} \cdot DR = \frac{CD}{R}$. When we let the index in our prescribed sum grow to ∞ , R also goes to ∞ , therefore we know that the two has the same limit since $\frac{CD}{R}$ goes to 0.

Problem 1.32. Using the notation from the previous problem, express $F_s(\omega_1, \omega_2)$ in terms of the values of $\wp(z)$ or a suitable derivative evaluated at values $z \in \Pi$ where $Nz \in L$.

Solution. Recall that we can again partition $\mathbb{Z} \times \mathbb{Z}$ to $N \times N$ blocks. Then any (m, n) can be written as:

$$(Na + r, Nb + t)$$

Then we know that when $s \geq 3$ the summation is absolutely convergent, thus can be rearranged and regrouped to:

$$\sum_{m,n} \frac{f(m,n)}{(m\omega_1 + n\omega_2)^s} = \frac{1}{N^s} \sum_{r,t \in \mathbb{Z}/N} \sum_{a,b \in \mathbb{Z}} \frac{f(r,t)}{(a\omega_1 + b\omega_2 + \frac{r}{N}\omega_1 + \frac{t}{N}\omega_2)^s}$$

Let's write $z_{r,t} = \frac{r}{N}\omega_1 + \frac{t}{N}\omega_2$, we know that $Nz_{r,t} \in L$, therefore $z_{r,t}$ is a N -torsion point of $\mathbb{C}/L$. Then we know that the sum can again be written as:

$$\frac{1}{N^s} \sum_{r,s \in \mathbb{Z}/N} \sum_{\omega \in \Omega} \frac{f(r,s)}{\omega + z_{r,t}}$$

Recall that for $s \geq 3$ we have:

$$\wp^{(s-2)}(z) = (-1)^{s-2}(s-1)! \sum_{\omega \in \Omega} \frac{1}{(z - \omega)^s}$$

Therefore we know that the above summation can be written as:

$$\frac{(-1)^{s-2}}{N^s(s-1)!} \sum_{r,s \in \mathbb{Z}/N} f(r,t) \wp^{(s-2)}(z_{r,t})$$

Now for $s = 2$, we don't have absolute convergence. Therefore we should be very careful when we try to regroup. Instead of doing the above rearranging argument, we consider the radius sum $S(R)$ we defined in the previous problem and we already know that this limit is equal to the sum with the prescribed order. Then recall that:

$$S(R) = \sum_{0 < |\lambda_{m,n}| \leq R} \frac{f(m,n)}{\lambda_{m,n}^2}$$

Notice that for any finite R , this sum is a finite sum, therefore it makes sense here to do any rearrangement we would like, which means we can write it as:

$$S(R) = \frac{1}{N^2} \sum_{r,t \in \mathbb{Z}/N} \sum_{|a| \leq \frac{R}{N}, |b| \leq \frac{R}{N}} f(r,t) \frac{1}{(a\omega_1 + b\omega_2 + z_{r,t})^2} = \frac{1}{N^2} \sum_{r,t \in \mathbb{Z}/N} \sum_{\omega \in \Omega, |\omega + z_{r,t}| \leq R/N} \frac{f(r,t)}{(\omega + z_{r,t})^2}$$

Let's pick out the term:

$$\sum_{\omega \in \Omega, |\omega + z_{r,t}| \leq R/N} \frac{f(r,t)}{(\omega + z_{r,t})^2}$$

Instead of using R/N , let's use T and we know that as $R \rightarrow \infty$ we also have $T \rightarrow \infty$. We will denote it by:

$$A_T(z) = \sum_{\omega \in \Omega, |\omega+z| \leq T} \frac{1}{(\omega+z)^2}$$

Recall that we have:

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in \Omega^*} \frac{1}{(z-\omega)^2} - \frac{1}{\omega^2}$$

What is important here is that if we denote B_T by:

$$B_T = \sum_{\omega \in \Omega^*, |\omega| \leq T} \frac{1}{|\omega|^2}$$

We know that:

$$A_T(z) - B_T = \sum_{\omega \in \Omega, |\omega+z| \leq T} \frac{1}{(\omega+z)^2} - \sum_{\omega \in \Omega^*, |\omega| \leq T} \frac{1}{|\omega|^2}$$

Notice that for a fixed z , when T grows to ∞ , $\sum_{\omega \in \Omega, |\omega+z| \leq T} \frac{1}{(\omega+z)^2}$ and $\sum_{\omega \in \Omega, |\omega| \leq T} \frac{1}{(\omega+z)^2}$ only differs by some elements in an annulus of fixed length. Then roughly, we have T times some constant elements, but notice that each of them only contributes $\frac{1}{T^2}$ to the sum, therefore we know that the difference should be something of the form $\frac{1}{T}$, therefore when T goes to infinity, it goes to 0, therefore we know that:

$$A_T(z) - B(T) = \wp(z) + o(1)$$

Therefore we know that:

$$S_R(z) = \frac{1}{N^2} \sum_{r,t \in \mathbb{Z}/N} A_T(z_{r,t}) \Rightarrow S_R(z) = \frac{1}{N^2} \sum_{r,t \in \mathbb{Z}/N} B_T + \wp(z_{r,t}) + o(1)$$

the last equation holds when we let $T \rightarrow \infty$ or in other words let $R \rightarrow \infty$. Then we know that:

$$\sum_{m,n} \frac{f(m,n)}{|m\omega_1 + n\omega_2|^2} = \lim_{R \rightarrow \infty} S_R = \lim_{T \rightarrow \infty} \frac{1}{N^2} f(r,t) \sum_{r,t \in \mathbb{Z}/N} B_T + \wp(z_{r,t}) + o(1)$$

Recall that since for any T , we have $\sum_{r,t \in \mathbb{Z}/N} f(r,t) B_T = 0$, we can get rid of that term and get:

$$\sum_{m,n} \frac{f(m,n)}{|m\omega_1 + n\omega_2|^2} = \lim_{R \rightarrow \infty} S_R = \lim_{T \rightarrow \infty} \frac{1}{N^2} f(r,t) \sum_{r,t \in \mathbb{Z}/N} \wp(z_{r,t}) + o(1)$$

Since $\sum_{r,t \in \mathbb{Z}/N}$ is a finite sum, we can put the limit inside and get:

$$\sum_{m,n} \frac{f(m,n)}{|m\omega_1 + n\omega_2|^2} = \frac{1}{N^2} \sum_{r,t \in \mathbb{Z}/N} f(r,t) (\lim_{T \rightarrow \infty} \wp(z_{r,t}) + o(1)) = \frac{1}{N^2} \sum_{r,t \in \mathbb{Z}/N} f(r,t) \wp(z_{r,t})$$

Problem 1.33. What can you say about how $\wp''(z)$ can be expressed in terms of $\wp(z)$.

Solution. Clearly, we know that $\wp''(z)$ is even, thus there should be a rational function in $\wp(z)$ representing $\wp''(z)$. We can actually say more, the pole of $\wp''(z)$ clearly is order 4.

Notice that we can choose $c \in \mathbb{C}$ such that;

$$\wp''(z) - c$$

has zeros at points not congruent to the half lattice points. We can also assume that they are all simple zeros, then we can assume they are $\{a_1, -a_1, a_2, -a_2\}$ which implies:

$$\wp''(z) - c = \lambda(\wp(z) - \wp(a_1))(\wp(z) - \wp(a_2))$$

where λ is a nonzero complex number since now they have the same zeros and poles.

1.6 Elliptic Functions in Weierstrass Form

As we discussed at the end of the last section, the square of $\wp'(z)$ is a cubic polynomial in $\wp(z)$. More precisely, we already saw that:

$$\wp'(z)^2 = C(\wp(z) - e_1)(\wp(z) - e_2)(\wp(z) - e_3)$$

where C is some constant. It's easy to find this constant C by comparing the coefficients of the lowest power of z in the Laurent expansion at the origin. Recall that by our definition of $\wp(z)$ and $\wp'(z)$, we know that:

$$\wp(z) - \frac{1}{z^2}, \quad \wp'(z) + 2\frac{1}{z^3}$$

are continuous at the origin. Therefore we know that the lowest term of $\wp'(z)^2$ is going to be $\frac{4}{z^6}$ and the lowest term of the right hand side is $C\frac{1}{z^6}$, therefore we conclude that $C = 4$. This means that if we consider the polynomial $f(x) = 4(x - e_1)(x - e_2)(x - e_3)$, then the Weierstrass p function and its derivative satisfies the following differential equation:

$$\wp'(z)^2 = f(\wp(z))$$

Recall that we already remarked that e_1, e_2, e_3 are all distinct in the previous.

Now I will refer you again to go to the algebraic geometry on compact Riemann surface for another way to derive this differential equation and what you will see there is:

$$(\wp'(z))^2 = 4\wp(z)^3 - g_2\wp(z) - g_3 \in \mathbb{C}[x]$$

where $g_2 = 60G_4$ and $g_3 = 140G_6$ with $G_4 = \sum_{l \in L^*} \frac{1}{l^4}$ and $G_6 = \sum_{l \in L^*} \frac{1}{l^6}$. We will have more exercises on this differential equation later.

For now, we want to focus on the fact that this differential equation actually has a very elegant and basic geometry interpretation. Consider the function:

$$\mathbb{C}/L \rightarrow \mathbb{P}_{\mathbb{C}}^2, \quad [z] \mapsto (\wp(z), \wp'(z), 1), \quad \forall z \neq 0$$

and we define $[0] \mapsto [0 : 1 : 0]$. Notice that this is indeed well defined precisely because $\wp, \wp'$ are elliptic. Notice that a nonzero point z is mapped to the xy -plane in the projective space with complex coordinates and notice that the x, y coordinate satisfies the equation:

$$y^2 = f(x), \quad f(x) = 4x^3 - g_2x - g_3$$

We already know that $f(x)$ is a cubic polynomial with distinct roots in $\mathbb{C}$. Therefore we know that every point on $\mathbb{C}/L$ is mapped to a point on the elliptic curve $y^2 = f(x)$ on $\mathbb{P}_{\mathbb{C}}^2$. Besides this the point $z = 0$ is mapped to the infinity point on the elliptic curve.

This map is one to one. To see this, let's first consider the points on the xy -plane. If $(x, y, 1)$ is such a point with $y^2 = f(x)$. We know that for all x except $x = e_1, e_2, e_3$, $\wp(z) = x$ has 2 distinct solutions in $\mathbb{C}/L$ and those two solutions z_1, z_2 gives you two values of $\wp'(z)$, which are the square roots of $f(x)$. So we know that each fixed $(x, y, 1)$ with $x \neq e_i$, it only has one element $z \in \mathbb{C}/L$ in the preimage. If x now is e_i , which means that $y = 0$, then we know that if $\wp'(z) = 0$, we know that there is only one z in $\mathbb{C}/L$ such that $\wp'(z) = 0$ and $\wp(z) = e_i$, i.e. the mid points, therefore we are done with the bijection at the xy -plane. What is left in the codomain is $(0 : 1 : 0)$ and what is left in the domain is 0 , and by definition, they match, thus we indeed get a bijection.

Moreover, this bijection is analytic in some sense which can be best described by using the notion of Riemann surfaces. To get a precise definition, I will again refer you to go to the notes on Riemann surfaces. But informally speaking, if you look at the map:

$$z \mapsto (\wp(z), \wp'(z), 1)$$

Away from 0 , we know that $\wp(z)$ and $\wp'(z)$ are holomorphic. Near 0 , the function looks like:

$$z \mapsto \left(\frac{\wp(z)}{\wp'(z)}, 1, \frac{1}{\wp'(z)} \right)$$

Since we know that $\wp'(z)$ has a higher order of pole at 0, then this is still holomorphic.

We want to conclude this section with a few remarks on the algebraic picture of the above geometric picture. First, notice that if we consider the polynomial $f(x) = y^2 \in \mathbb{C}[x, y]$, we know that if we view it as an element in $\mathbb{C}[x][y]$, then by Gauss's lemma, we know that $y^2 - f(x)$ is irreducible iff it's irreducible in $\mathbb{C}(x)[y]$. Since in $\mathbb{C}(x)[y]$, the polynomial is only degree 2, therefore it's not irreducible iff there is some element g in $\mathbb{C}[x]$ such that:

$$f(x) = g^2$$

This is impossible since $f(x)$ is degree 3, which is odd. Therefore $f(x, y) \in \mathbb{C}[x, y]$ is irreducible, thus defines a principal prime ideal. In algebraic geometry, there is an object which is called a closed subscheme of $\mathbb{P}_{\mathbb{C}}^2$ defined by the polynomial $y^2 - f(x)$. In the xy -plane of $\mathbb{P}_{\mathbb{C}}^2$, it looks like the curve $f(x) = y^2$ on the plane. This is an irreducible and reduced closed subvariety of $\mathbb{A}_{\mathbb{C}}^2$ and we can compute it's function field $K(E) = \text{Frac}(\mathbb{C}[x, y]/(y^2 - f(x)))$ which is isomorphism to $\mathbb{C}(x)[y]/(y^2 - f(x))$ via the following diagram:

$$\begin{array}{ccc} & \mathbb{C}[x, y]/(y^2 - f(x)) & \\ \swarrow & & \searrow \\ \text{Frac}(\mathbb{C}[x, y]/(y^2 - f(x))) & \xleftarrow{\quad} & \mathbb{C}(x)[y]/(y^2 - f(x)) \end{array}$$

This function field can be basically thought as the field of all meromorphic functions on E , or at least all the meromorphic functions of E lying inside the xy -plane. In particular, we see that the degree of the field extension $[\mathbb{C}(x)[y]/(y^2 - f(x)) : \mathbb{C}(x)]$ is 2, thus every element in $\mathbb{C}(x)[y]/(y^2 - f(x))$ is of the form $f(x) + g(x)y$ where $f(x), g(x)$ are rational functions in $\mathbb{C}$.

We will also have a map of algebraic varieties:

$$\mathbb{C}/L \rightarrow E$$

which basically is the analytic map we wrote down before, and on the level of function field, or the field of meromorphic functions. What the above map of algebraic varieties does on the field of meromorphic functions is that given a meromorphic function $f(x)[y]$ in $\mathbb{C}(x)[y]/(y^2 - f(x))$, it will map it to a meromorphic function on $\mathbb{C}/L$ as the map is defined to be:

$$f(x)[y] \mapsto f(\wp(z))\wp'(z)$$

This is a well defined map since we know that $y^2 - f(x)$ is mapped to $(\wp'(z))^2 - f(\wp(z)) = 0$. This is actually an isomorphism of fields since we know that each meromorphic function of $\mathbb{C}/L$ is of the form:

$$f(\wp(z)) + \wp'(z)g(\wp(z)), f, g \in \mathbb{C}(x)$$

and we know that elements in $\mathbb{C}(x)[y]/(y^2 - f(x))$ are also of this form. This shows surjectivity. The fact it's injective just comes from the fact that it's a map of rings between fields. This also shows that there are no further relations between $\wp(z), \wp'(z)$ rather than the one we already found.

You can also try to make the ramification of the elliptic curves more algebraic but let's just stop here as we will not in this notes use any algebraic geometric techniques later on.

Below are the exercises of this section:

Problem 1.34. Show that for the lattice of Gaussian integers $\mathbb{Z}[i]$ g_2 is a nonzero real number and $g_3 = 0$.

Solution. Recall that:

$$G_6 = \sum_{\omega \in \Omega^*} \frac{1}{\omega^6}$$

Notice that for this special lattice multiply by i fixes the lattice, then using the fact that the series converges uniformly, we can use any order of summation, therefore we know that:

$$G_6 = \sum_{\omega \in \Omega^*} \frac{1}{(i\omega)^6} = - \sum_{\omega \in \Omega}^* \frac{1}{\omega^6} \Rightarrow \sum_{\omega \in \Omega}^* \frac{1}{\omega^6} = 0$$

Similarly, notice that complex conjugation also preserves the lattice therefore:

$$G_4 = \overline{G_4} \Rightarrow G_4 \in \mathbb{R}$$

To show it's nonzero, recall that the discriminant of the elliptic curve $4x^2 - g_2x - g_3$ is given by:

$$\Delta = g_2^3 - 27g_3^2 = g_2^2 \neq 0$$

Therefore g_2^3 should not be 0 which implies that g_2 is not 0.

1.7 Addition Law

In the last section, we showed how the Weierstrass $\wp(z)$ function gives a correspondence between $\mathbb{C}/L$ and the points on the elliptic curve $E : y^2 = f(x) = 4x^3 - g_2(L)x - g_3$ in $\mathbb{P}_{\mathbb{C}}^2$.

Note that we have an obvious addition on $\mathbb{C}/L$, which is just the addition coming from $\mathbb{C}$ but modulo L . Then a natural thought would be to use the addition relation on $\mathbb{C}/L$ to define an addition relations for the points in E . What we do is that given two points P_1, P_2 on E , we then go back to the complex number plane to find z_1, z_2 such that P_1 corresponds to z_1 and P_2 correspond to z_2 , we will define $P_1 + P_2$ to be the point corresponding to $z_1 + z_2$. Here Notice that this in general is just very routine since given a bijection between an additive group and another set, we can always define an additive groups structure on the set via the bijection.

However, what is remarkable about this construction here is that the addition law we obtained can be geometrically interpreted in a very simple term. The purpose of this section is to show how this is done.

We will use the following lemma whose proof you can find in the complex algebraic geometry notes:

Lemma 1.35. Let f be an elliptic function with respect to L and let Π be the fundamental parallelogram of L , let's choose α small such that the translation $\alpha + \Pi$ has no zeros and poles on the boundary, then suppose that $a_1, \dots, a_r$ are the zeros with multiplicity $k_1, \dots, k_r$. $b_1, \dots, b_s$ are the poles with multiplicity $l_1, \dots, l_s$, then we have:

$$\sum_i a_i k_i - \sum_j b_j l_j \in L$$

Using this result, let $z \in \mathbb{C}/L$ be a point, we will let P_z be the corresponding point on the elliptic curve. We already know that if $z = 0$, then the point is $[0 : 1 : 0]$ and if z is not 0, we know that the point $P_z = [\wp(z), \wp'(z) : 1]$. Suppose that we want to add P_{z_1} and P_{z_2} to obtained $P_{z_1+z_2}$. Clearly, the additive identity would be $P_0 = [0 : 1 : 0]$. Then we are interested what is the additive inverse of a nonzero element. Notice that the nonzero points on the curve are actually given on the xy -plane of the projective space and their coordinates are always $[x : y : 1]$.

Now given $P_{z_1} = [x_1 : y_1 : 1]$ and $P_{z_2} = [x_2 : y_2 : 1]$ and assume that they have the same x value but not the same point. This means that $x_1 = x_2$ and $y_1 \neq y_2$, which further implies that $\wp(z_1) = \wp(z_2) = x_1$ but $z_1 \neq z_2$. Since we know that the order of $\wp(z)$ is 2 and $z_1 \neq z_2$, then we know that x_1 is not e_1, e_2, e_3 . Then we know that in this case $\wp(z_1) = \wp(-z_1)$ and $z_1 \neq -z_1$, therefore $z_2 = -z_1$ and $\wp'(z_2) = \wp'(-z_1) = -\wp'(z_1)$, thus $y_2 = -y_1$. In this case we know that $P_{z_1} + P_{z_2} = P_{z_1+(-z_1)} = P_0$. Now if $y = 0$, then we know that $x_1 = e_i$ which means that z is a half lattice point, which means that $z + z = 0$, therefore we know a point is on the x -axis, then the additive inverse is itself, i.e. $[x : 0 : 1]$. Geometrically speaking, we see that the two points are on the same vertical line in the xy -plane, it's just sometimes, the tangent line intersects the curve at two distinct points, but other times, the line only intersect at one point, but with multiplicity 2. Then we have proved the following:

Proposition 1.36. The additive inverse of $[x, y : 1]$ is $[x, -y : 1]$.

What is interesting is that if we consider the vertical line $x = a$ in the xy -plane. Then we find that the corresponding line in the projective plane is $x = az$, which means that it passes through the point at infinity $[0 : 1 : 0]$. Therefore if we ask how many points does the line joining $[x : y : 1]$ and $[x : -y : 1]$, the answer is always three points including the point at infinity and counting multiplicity.

Now given P_1, P_2 on the elliptic curve and assume that neither of them is the point at infinity, then we know that there is a unique line $l = \overline{P_1 P_2}$ joining them. If $P_1 = P_2$, let's just take the tangent line at that point. If l happens to be a vertical line, i.e. a line whose equation has no y terms. Then we already see that $P_1 + P_2 = 0$. If l is not vertical, we want to use the line to find what is $P_1 + P_2 = P_3$. First I claim that besides P_1, P_2 , the line has another intersection point with the curve which we call P'_3 . Indeed, if the line is of the form $y = mx + b$, then in order to compute the intersection point $[x : y : 1]$, we set:

$$(mx + b)^2 = 4x^3 - g_2x - g_3$$

Notice that this is a cubic polynomial, thus always comes with 3 solutions in $\mathbb{C}$. Each of which gives a point of intersection. If x is a double root or triple root, then l intersects with the curve with multiplicity 2 or 3 at the point (x, y) . In any case, the total numbers of points is 3. Notice that the vertical line also intersects the curve at three points, counting the point at infinity. Moreover, if we consider the line at infinity $z = 0$, this also intersects with the curve at three points. Indeed, we know that the line is tangent to this curve and its equation is $z = 0$, then if we consider the multiplicity of the intersection, we can look at in the xz plane, where the equation becomes:

$$\frac{z^2}{y} = \frac{x^3}{y^3} - g_2 \frac{x}{y} \frac{z^2}{y^2} - g_3 \frac{z^3}{y^3}$$

Then if we plug $\frac{z}{y} = 0$, then it becomes:

$$0 = \frac{x^3}{y^3}$$

Clearly the multiplicity is 3 at $\frac{x}{y} = 0$. In conclusion this is just a special case of the following theorem:

Theorem 1.37. (Bezout's Theorem) Let $\tilde{F}(x, y, z), \tilde{G}(x, y, z)$ be homogeneous polynomials of degree m, n respectively over an algebraically closed field K , suppose that $\tilde{F}$ and $\tilde{G}$ has no common polynomial factors, then the two curves in $\mathbb{P}_K^2$ intersects at mn points counting multiplicity.

We won't prove it here and you can go to the series on FOAG for details.

Proposition 1.38. If $P_1 + P_2 = P_3$, then $-P_3$ is the third point of intersection of the line $l = \overline{P_1 P_2}$ with the elliptic curve. If $P_1 = P_2$, then by $\overline{P_1 P_2}$, we just mean the tangent line at P_1 .

Proof. Recall that if P_1, P_2 are both $[0 : 1 : 0]$, we already saw that the intersection point has multiplicity 3 at $[0 : 1 : 0]$. If one of them is infinity and another one of them is on the xy -plane, we know that the line passing $[0 : 1 : 0]$ is $ax + cz = 0$ and if $a = 0$, we know that it becomes $z = 0$, which contradicts to the fact that some $[x : y : 1]$ is on the line. Then we know that in this case, this becomes the vertical line in the xy -plane, then we know that in this case if P_2 is the point in the xy -plane, another the intersection point just $-P_2$.

Finally, let's assume that P_1, P_2 are all in xy -plane. Then we assume that the line passing them is $y = mx + b$, and we know that if we want to say that a point $P_z = [\wp(z), \wp'(z) : 1]$ lies in the line means that:

$$\wp'(z) - m\wp(z) - b = 0$$

Notice that this is again an elliptic function of order 3 therefore it should have three zeros counted with multiplicity. We already know that z_1, z_2 are zeros, therefore we know that there is another zero z_3 , and from the lemma in the beginning, we have:

$$\sum_i z_1 + z_2 + z_3 - 0 \in L$$

That is to say that the class of $[z_3]$ is going to be $-[z_1 + z_2]$. Therefore the point it correspond on the curve is:

$$P_{-(z_1+z_2)} = -P_{z_3}$$

However, there is a problem in the above argument. To see what the problem could be, we ask that if we have a triple zero of the solution:

$$\wp'(z) - m\wp(z) - b = 0$$

In this case, let's write $[z_1] = [z_2] = [z_3]$, and we know that the three points correspond to the same point on the elliptic curve. But the question is that does this point on the curve has multiplicity 3. In other words, we ask does the multiplicity of roots of an elliptic function agree with the multiplicity of the intersection on the elliptic curve.

The proof of this is a bit ad hoc. First consider the classes $[z_1]$, $[z_2]$ and $[-z_3]$ which are three zeros of the $\wp'(z) - m\wp(z) - b$ listed with multiplicity, we know that we can assume that $[-z_i]$ is not $[z_j]$ for $i \neq 0$. Otherwise, we will get that the line is vertical. Then since $[-z_1]$, $[-z_2]$ and $[z_3]$ are the roots for:

$$\wp'(z) + m\wp(z) + b = 0$$

Then we know that the 6 zeros of $\wp'(z)^2 - (m\wp(z) + b)^2 = f(\wp(z)) - (m\wp(z) + b)^2 = 0$ are given by $\pm[z_1]$, $\pm[z_2]$ and $\pm[z_3]$. Let x_1, x_2, x_3 be the three solutions of $f(\wp(z)) - (m\wp(z) + b)^2 = 0$, which means that we have the factorization:

$$f(\wp(z)) - (m\wp(z) + b)^2 = 4(\wp(z) - x_1)(\wp(z) - x_2)(\wp(z) - x_3)$$

If say $\wp(-z_1) = \wp(z_1) = x_1$, $\wp(-z_2) = \wp(z_2) = x_2$ and $\wp(z_3) = \wp(-z_3) = x_3$, then the multiplicity of x_1 depends on how many equalities of the form:

$$\wp(z_1) = \wp(z_2), \quad \wp(z_1) = \wp(z_3)$$

However, using the property of $\wp(z)$, we know that $\wp(x) = \wp(y)$ iff $\pm x = \pm y$, therefore the multiplicity depends on how many $\pm z_2, \pm z_3$ are equal to z_1 or $-z_1$. However, we know that $-z_2 \neq z_1$ and $z_3 \neq z_1$, therefore $z_2 \neq -z_1$ and $-z_3 \neq -z_1$. Therefore it just depends the number of "true" equalities below:

$$z_2 = z_1, \quad -z_3 = z_1$$

(Since the other two equalities we have $-z_2 = -z_1$ and $z_3 = -z_1$ are equivalent to the equations above.) Therefore the notion of intersection multiplicity is the same as the multiplicity of the order of the function $\wp'(z)^2 - m\wp(z) - b = 0$. $\square$

We could have establish the the group law in this geometric form in the beginning and then prove it's a group law and commutative. It turns out that the most difficult part is to prove associativity which requires further study which we don't do.

Note that even though the above proof using $\wp(z)$ is very good, a priori it can be only used to show the group law for elliptic curves of the form $y^2 = 4x^2 - g_2(L)x - g_3(L)$ for some lattice L . We will later show that if an elliptic curved can be transformed to this equation using some linear change of variables, then the above discuss is still valid. As what we have mentioned, we will also prove later that at least over the complex numbers, any elliptic curve can be transformed to to Weierstrass form for some lattice L .

For example, if we consider the lattice of the Gaussian integers $\mathbb{Z}[i]$, then I claim that g_2 for this lattice is a positive real number and g_3 for this lattice is 0. Indeed, recall that g_2 is defined by $60G_4$ and g_3 is defined by $140G_6$. Recall that we already proved that

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